Chapter 1 : Sequences and Series

EXERCISE 1

1. Write down the first four terms of the sequence if you start with :

a. 4 and add 9 each time

b. 45 and subtract 6 each time

c. 2 and multiply by 3 each time

d. 96 and divide by 2 each time

2. For each of the following write a description of the sequence and find the next 2 terms :

a. 8, 16, 24, 32, ...

b. 2, 5, 8, 11, ...

c. 36, 31, 26, 21, ...

d. 96, 89, 82, 75, ...

e. 1, 4, 16, 64, ...

f. 2, 6, 18, 54, ...

g. 480, 240, 120, 60, ...

h. 243, 81, 27, 9, ...

i. 50 000, 10 000, 2000, 400, ...

3. Describe the following number patterns and write down the next 3 terms :

a. 1, 4, 9, 16, ...

b. 1, 8, 27, 64, ...

c. 2, 6, 12, 20, ...

4. Find the next two terms of :

a. 95, 91, 87, 83, ...

b. 5, 20, 80, 320, ...

c. 1, 16, 81, 256, ...

d. 2, 3, 5, 7, 11, ...

e. 2, 4, 7, 11, ...

f. 9, 8, 10, 7, 11, ...

EXERCISE 2

1. Write down the first four terms in each sequence.

a. $u_1 = 10, \quad u_{n+1} = 3(u_n)$

b. $u_1 = 3, \quad u_{n+1} = 2u_n + 1$

c. $u_1 = \frac{3}{4}, \quad u_{n+1} = \frac{2}{3}(u_n)$

d. $u_1 = x, \quad u_{n+1} = (u_n)^2$

2. Write a recursive formula for each sequence.

a. $2, 4, 6, 8, \ldots$

b. $1, 3, 9, 27, \ldots$

c. $64, 32, 16, 8, \ldots$

d. $7, 12, 17, 22, \ldots$

EXERCISE 3

1. Write down the first four terms in each sequence.
   - **a.** $u_n = 3^n$
   - **b.** $u_n = -6n + 3$
   - **c.** $u_n = 2^{n-1}$
   - **d.** $u_n = n^n$
2. Write a general formula for the nth term of each sequence.
   - **a.** $2, 4, 6, 8, \ldots$
   - **b.** $1, 3, 9, 27, \ldots$
   - **c.** $64, 32, 16, 8, \ldots$
   - **d.** $7, 12, 17, 22, \ldots$
   - **e.** $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \ldots$
   - **f.** $x, 2x, 3x, 4x, \ldots$
3. The sequence $1, 1, 2, 3, 5, 8, 13, \ldots$ is known as the Fibonacci sequence.
   - **a.** Find the 15th term of the Fibonacci sequence.
   - **b.** Write a recursive formula for the Fibonacci sequence.

EXERCISE 4

1. An arithmetic sequence has first term 19 and 15th term 31.6. Find the common difference.
2. In an arithmetic sequence, $u_{10} = 37$ and $u_{21} = 4$. Find the common difference and the first term.
3. Find the value of x in the arithmetic sequence $3, x, 8, \ldots$
4. Find the value of m in the arithmetic sequence $m, 13, 3m - 6, \ldots$

EXERCISE 5 A sequence is defined by

$$u_n = 3n - 2.$$

1. Prove that the sequence is arithmetic. Hint : Find $u_{n+1} - u_n$.
2. Find u_1 and d.
3. Find the 57th term.
4. What is the largest term of the sequence that is smaller than 450 ? Which term is this ?

EXERCISE 6

€1000 is invested in an account paying 6% compound interest per annum. Calculate the amount of money in the account after 10 years if compounding is :

1. annual
2. quarterly
3. monthly

1. Victoria put €1500 into a savings account at 4% annual interest compounded semiannually. Find the value after 10 years.
2. At the birth of her daughter Jane, Charlotte deposited £500 into a savings account. The annual interest rate was 4% compounded quarterly . How much money will Jane have on her 16th birthday ?
3. How much money should be invested now if you wish to have €4000 in your account after 6 years if interest is compounded quarterly at an annual rate of 5% ?
4. In 2017, the population of a town in Switzerland was estimated to be 7554. How large would the town's population be in 2022 if it grows at a rate of 0.5% annually ?
5. You invest €1000. What annual rate is needed to double it in 10 years if interest is compounded quarterly ?

Problem A farmer feeds his cattle herd with hay every day in July. The amount of hay in his barn at the end of day n is given by the arithmetic sequence

$$u_n = 100 - 2.7n \quad \text{tonnes}.$$

1. Write down the amount of hay in the barn on the first three days of July.
2. Find and interpret the common difference.
3. Find and interpret u_{25}.
4. How much hay is in the barn at the beginning of August ?

EXERCISE 7 A. Write an expression for each series using sigma notation.

1. $1+2+3+4+5+6+7+8$
2. $9+16+25+36+49$
3. $27+25+23+21+19+17$
4. $240+120+60+30+15+7.5$
5. $5x+6x+7x+8x+9x+10x$
6. $4+7+10+13+\cdots+55$
7. $1+3+9+27+\cdots+59049$
8. $a+2a^2+3a^3+4a^4+5a^5$

B. Write each series as a sum of terms.

1. $\sum_{n=1}^{8}(3n+1)$
2. $\sum_{a=1}^{5}(4^a)$
3. $\sum_{r=3}^{7}(5\cdot 2^r)$
4. $\sum_{n=5}^{11}(x^n)$

C. Evaluate.

1. $\sum_{n=1}^{9}(8n-5)$
2. $\sum_{r=1}^{5}(3^r)$
3. $\sum_{m=1}^{7}(m^2)$
4. $\sum_{x=4}^{10}(7x-4)$

EXERCISE 8

A. For the following sequences :

i) Write down an expression for S_n.

ii) Find S_5.

1. $3, 11, 19, 27, \ldots$

2. $42, 37, 32, 27, \ldots$

3. $12, 6, 3, 1\frac{1}{2}, \ldots$

4. $2, 3, 4\frac{1}{2}, 6\frac{3}{4}, \ldots$

5. $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \ldots$

6. $1, 8, 27, 64, \ldots$

B. Expand and evaluate :

1. $\sum_{k=1}^{3} 4k$

2. $\sum_{k=1}^{6} (k+1)$

3. $\sum_{k=1}^{4} (3k-5)$

4. $\sum_{k=1}^{5} (11-2k)$

5. $\sum_{k=1}^{7} k(k+1)$

6. $\sum_{k=1}^{5} 10 \times 2^{k-1}$

C. For $u_n = 3n - 1$, write using sigma notation and evaluate the sum :

$$u_1 + u_2 + u_3 + \cdots + u_{20}$$

D. Show that :

1. $\sum_{k=1}^{n} c = cn$

2. $\sum_{k=1}^{n} ca_k = c\sum_{k=1}^{n} a_k$

3. $\sum_{k=1}^{n} (a_k + b_k) = \sum_{k=1}^{n} a_k + \sum_{k=1}^{n} b_k$

E.

1. Expand $\sum_{k=1}^{n} k$.

2. Now write the sum with terms in the reverse order, placing each term under a term in the original expansion. Add each term with the one under it.

3. Hence write an expression for the sum S_n of the first n integers.

4. Hence find a and b if

$$\sum_{k=1}^{n} (ak + b) = 8n^2 + 11n$$

for all positive integers n.

EXERCISE 9

1. Find the sum of :

1. $3 + 7 + 11 + 15 + \cdots$ to 20 terms

2. $\frac{1}{2} + 3 + \frac{5}{2} + 8 + \cdots$ to 50 terms

3. $100 + 93 + 86 + 79 + \cdots$ to 40 terms

4. $50 + 48\frac{1}{2} + 47 + 45\frac{1}{2} + \cdots$ to 80 terms

2. Find the sum of :

1. $5+8+11+14+\cdots+101$
2. $50+49\frac{1}{2}+49+48\frac{1}{2}+\cdots+(-20)$
3. $8+10\frac{1}{2}+13+15\frac{1}{2}+\cdots+83$

3. Evaluate these arithmetic series :

1. $\sum_{k=1}^{20}(2k+5)$
2. $\sum_{k=1}^{15}(k-50)$
3. $\sum_{k=1}^{20}\left(\frac{k+3}{2}\right)$

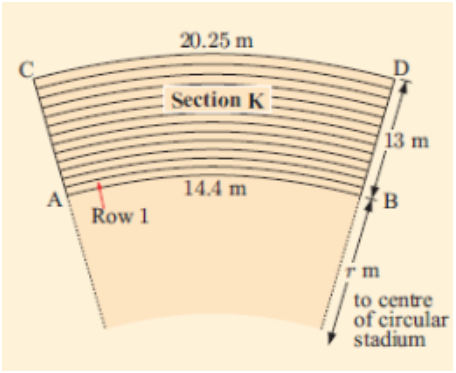
Hint : List the first 3 terms and the last term. Check your answers using technology.

4. An arithmetic series has seven terms. The first term is 5 and the last term is 53. Find the sum of the series.

5. An arithmetic series has eleven terms. The first term is 6 and the last term is -27. Find the sum of the series.

EXERCISE 10 Stadium Problem

A circular stadium consists of sections as illustrated, with aisles in between. The diagram shows the 13 tiers of concrete steps for the final section, Section K. Seats are placed along every concrete step, with each seat 0.45 m wide. The arc AB at the front of the first row is 14.4 m long, while the arc CD at the back of the back row is 20.25 m long.



1. How wide is each concrete step ?
2. What is the length of the arc of the back of Row 1, Row 2, Row 3, and so on ?
3. How many seats are there in Row 1, Row 2, Row 3, ..., Row 13 ?
4. How many sections are there in the stadium ?
5. What is the total seating capacity of the stadium ?
6. What is the radius r of the 'playing surface' ?

EXERCISE 11 Find the sum of the following series :

1. $12 + 6 + 3 + 1.5 + \cdots$ to 10 terms
2. $\sqrt{7} + 7 + 7\sqrt{7} + 49 + \cdots$ to 12 terms
3. $6 - 3 + \frac{3}{2} - \frac{3}{4} + \cdots$ to 15 terms
4. $1 - \frac{1}{\sqrt{2}} + \frac{1}{2} - \frac{1}{2\sqrt{2}} + \cdots$ to 20 terms

EXERCISE 12 Find a formula for S_n, the sum of the first n terms of the series :

1. $\sqrt{3} + 3 + 3\sqrt{3} + 9 + \cdots$
2. $12 + 6 + 3 + 1\frac{1}{2} + \cdots$
3. $0.9 + 0.09 + 0.009 + 0.0009 + \cdots$
4. $20 - 10 + 5 - 2\frac{1}{2} + \cdots$

EXERCISE 13 Evaluate these geometric series :

1. $\sum_{k=1}^{10} 3 \times 2^{k-1}$
2. $\sum_{k=1}^{12} \left(\frac{1}{2}\right)^{k-2}$
3. $\sum_{k=1}^{25} 6 \times (-2)^k$

EXERCISE 14 Financial Problem

Each year a salesperson is paid a bonus of \$2000 which is banked into the same account. It earns a fixed rate of interest of 6% p.a., with interest being paid annually. The total amount in the account at the end of each year is calculated as follows :

$$A_0 = 2000$$

$$A_1 = A_0 \times 1.06 + 2000$$

$$A_2 = A_1 \times 1.06 + 2000 \quad \text{and so on.}$$

1. Show that $A_2 = 2000 + 2000 \times 1.06 + 2000 \times (1.06)^2$.
2. Show that $A_3 = 2000[1 + 1.06 + (1.06)^2 + (1.06)^3]$.
3. Find the total bank balance after 10 years, assuming there are no fees or withdrawals.

EXERCISE 15 Consider

$$0.\overline{3} = \frac{3}{10} + \frac{3}{100} + \frac{3}{1000} + \cdots$$

which is an infinite geometric series.

1. Find :
   a. u_1
   b. r
2. Using a, show that $0.\overline{3} = \frac{1}{3}$.

EXERCISE 16 Write as a rational number :

1. $0.\overline{7}$
2. $0.\overline{16}$
3. $0.\overline{312}$

EXERCISE 17 Find the sum of each of the following infinite geometric series :

1. $18 + 12 + 8 + \frac{16}{3} + \cdots$
2. $18.9 - 6.3 + 2.1 - 0.7 + \cdots$

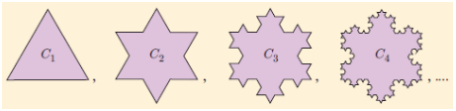
EXERCISE 18 Find each of the following :

1. $\sum_{k=1}^{\infty} \frac{3}{4^k}$
2. $\sum_{k=0}^{\infty} 6\left(-\frac{2}{5}\right)^k$

EXERCISE 19 The sum of the first three terms of a convergent infinite geometric series is 19. The sum of the series is 27. Find the first term and the common ratio.

EXERCISE 20 Von Koch's Snowflake Curve

In this investigation we consider a *limit curve* named after the Swedish mathematician Niels Fabian Helge von Koch (1870–1924).



To draw Von Koch's Snowflake curve we :

- start with an equilateral triangle, C_1,
- then divide each side into 3 equal parts,
- then on each middle part draw an equilateral triangle,
- then delete the side of the smaller triangle which lies on C_1.

The resulting curve is C_2. By repeating this process on every edge of C_2, we generate curve C_3.

We hence obtain a sequence of special curves $C_1, C_2, C_3, C_4, \ldots$ and Von Koch's curve is the limiting case when n is infinitely large.

Your task is to investigate the perimeter and area of Von Koch's curve.

What to do :

1. Suppose C_1 has a perimeter of 3 units. Find the perimeter of C_2, C_3, C_4, and C_5.

Hint : a line segment divided into 3 parts  becomes 4 parts.

Remembering that Von Koch's curve is C_n, where n is infinitely large, find the perimeter of Von Koch's curve.

2. Suppose the area of C_1 is 1 unit^2. Explain why the areas of C_2, C_3, C_4, and C_5 are :

$$A_2 = 1 + \tfrac{1}{3} \quad \text{units}^2,$$

$$A_3 = 1 + \tfrac{1}{3}\left[1 + \tfrac{4}{9}\right] \quad \text{units}^2,$$

$$A_4 = 1 + \tfrac{1}{3}\left[1 + \tfrac{4}{9} + \left(\tfrac{4}{9}\right)^2\right] \quad \text{units}^2,$$

$$A_5 = 1 + \tfrac{1}{3}\left[1 + \tfrac{4}{9} + \left(\tfrac{4}{9}\right)^2 + \left(\tfrac{4}{9}\right)^3\right] \quad \text{units}^2.$$

Use your calculator to find A_n where $n = 1, 2, 3, 4, 5, 6, 7$, giving answers which are as accurate as your calculator permits.

What do you think will be the area within Von Koch's snowflake curve ?

3. Is there anything remarkable about your answers to 1 and 2 ?

4. Similarly, investigate the sequence of curves obtained by adding squares on successive curves from the middle third of each side. These are the curves $C_1, C_2, C_3, \ldots$ shown below.

